

ESSENTIALS OF CLOUD COMPUTING

Cloud Computing Paradigms

Unit 1: Evolution, Service Models, and Architectures

Based on K. Chandrasekaran

The Evolution of Computing



Computing has shifted from centralized mainframes to highly distributed, internet-based utility models.

Parallel vs. Distributed Systems



Parallel Computing

Multiple processors execute instructions simultaneously within a single machine to solve complex problems faster. Shared memory is common.



Distributed Computing

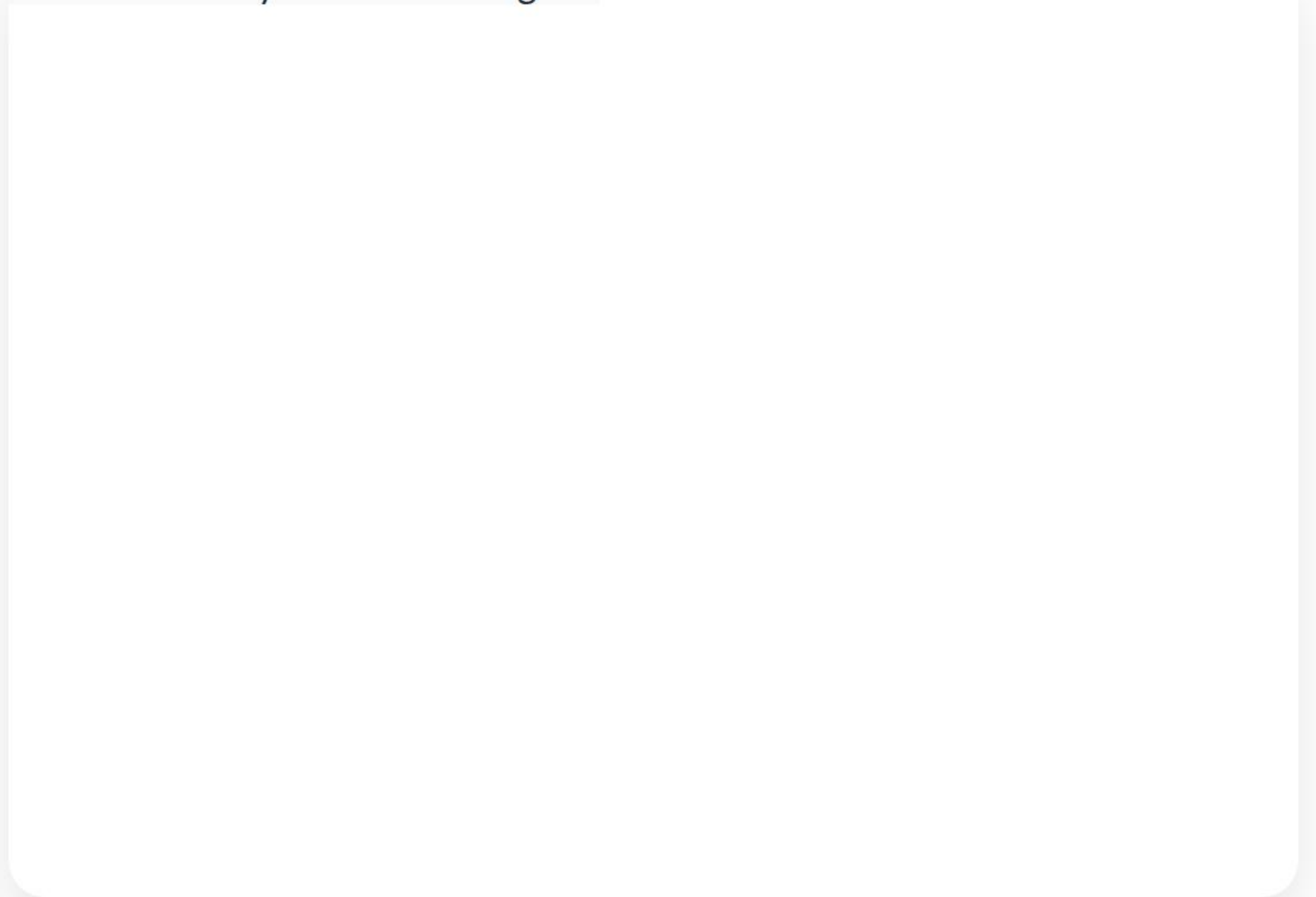
Independent computers connected via a network work as a single system. They use message passing and local memory to communicate.

Cluster Computing Paradigms

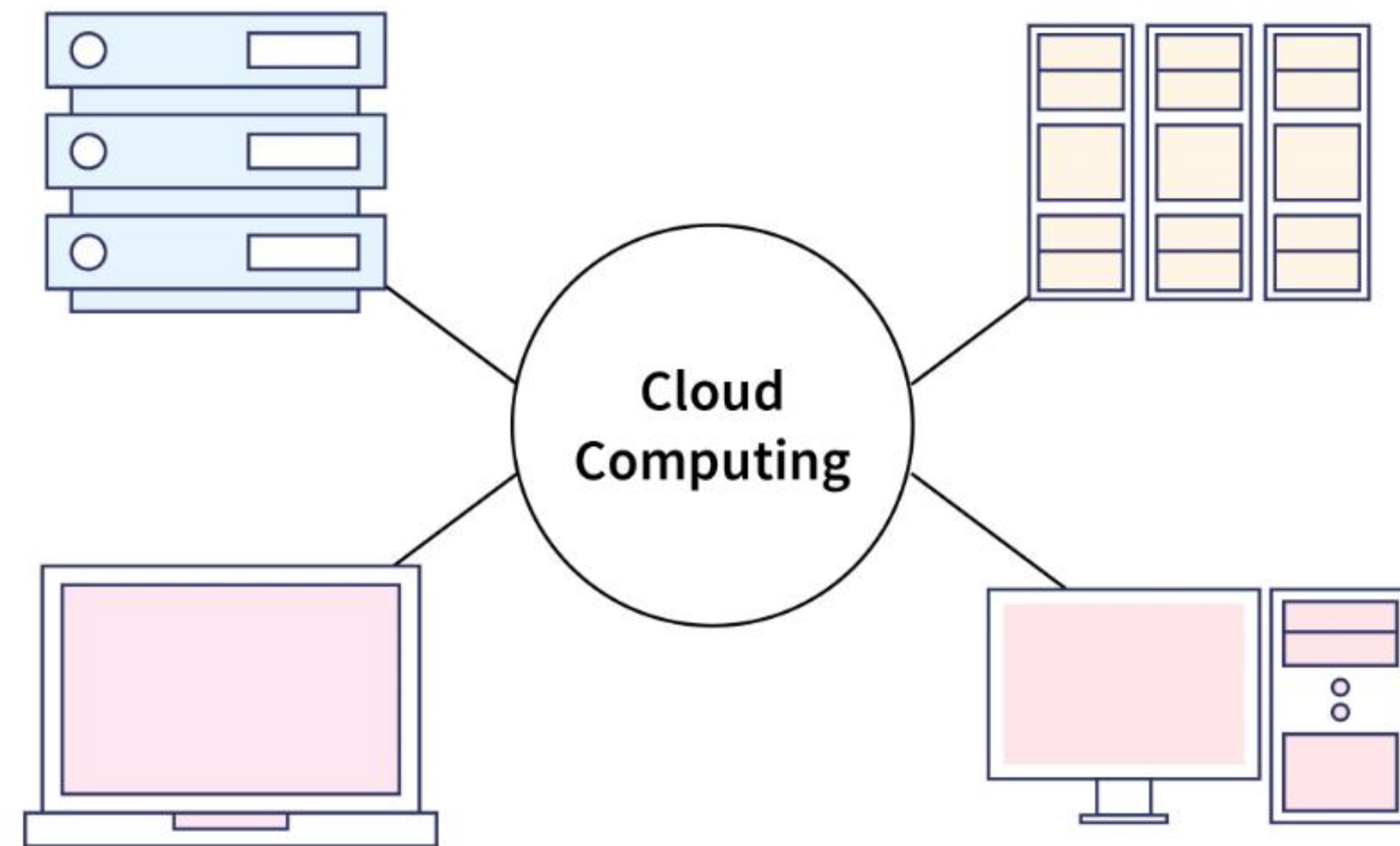
A cluster is a group of interconnected, independent computers (nodes) working together as a single, integrated resource.

- **High Performance:** Aggregates processing power.
- **High Availability:** Fault tolerance through redundancy.
- **Scalability:** Easy to add new nodes to the pool.

 Distributed system nodes diagram



Grid: The Virtual Supercomputer



SCALER
Topics

Federated Resources

Grid computing constitutes a network of computers working together to perform massive tasks that are too difficult for single machines.

- **Heterogeneous:** Mix of different hardware.
- **Geographical Spread:** Nodes can be anywhere.
- **Dynamic:** Nodes join and leave the "grid".

Utility Computing: Pay-as-you-go

100%

On-Demand Provisioning

Computing resources are treated like a public utility (like electricity or water). Users pay only for the resources consumed.

The business backbone of modern cloud computing services.

Edge and Fog Paradigms



Edge Computing

Brings computation to the network's edge (IoT devices, user PCs) to reduce latency and bandwidth usage.



Fog Computing

Acts as an intermediate structure between the cloud and data-producing devices (local LAN level).

NIST Definition

"Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort."

— National Institute of Standards and Technology (NIST)

5 Essential Characteristics

1. On-Demand Self-Service: Unilateral resource provisioning.

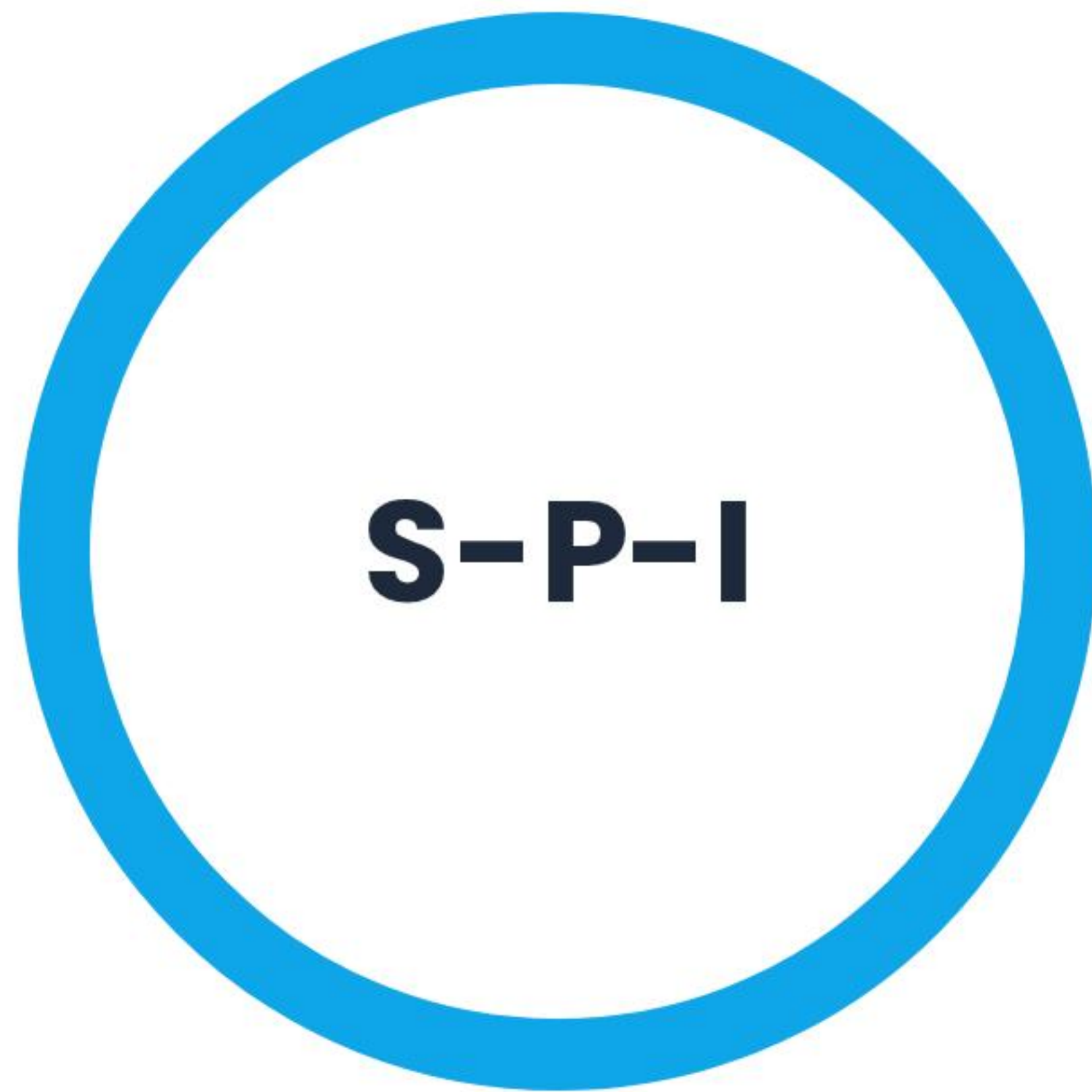
2. Broad Network Access: Accessible via standard mechanisms.

3. Resource Pooling: Multi-tenant model serving many users.

4. Rapid Elasticity: Scaling up or down instantly.

5. Measured Service: Automatic monitoring and metering of usage.

Service Models (The SPI Model)



SaaS | PaaS | IaaS

- ✓ SaaS: Software applications for end users.
- ✓ PaaS: Platforms for developers to build apps.
- ✓ IaaS: Fundamental infrastructure (Servers, Storage).

IaaS: Infrastructure

Infrastructure as a Service delivers fundamental computing, storage, and networking resources on demand.

- Virtualization technologies.
- Full control over the OS.
- Scalable storage (e.g., AWS S3).
- Networking (VPCs, Firewalls).



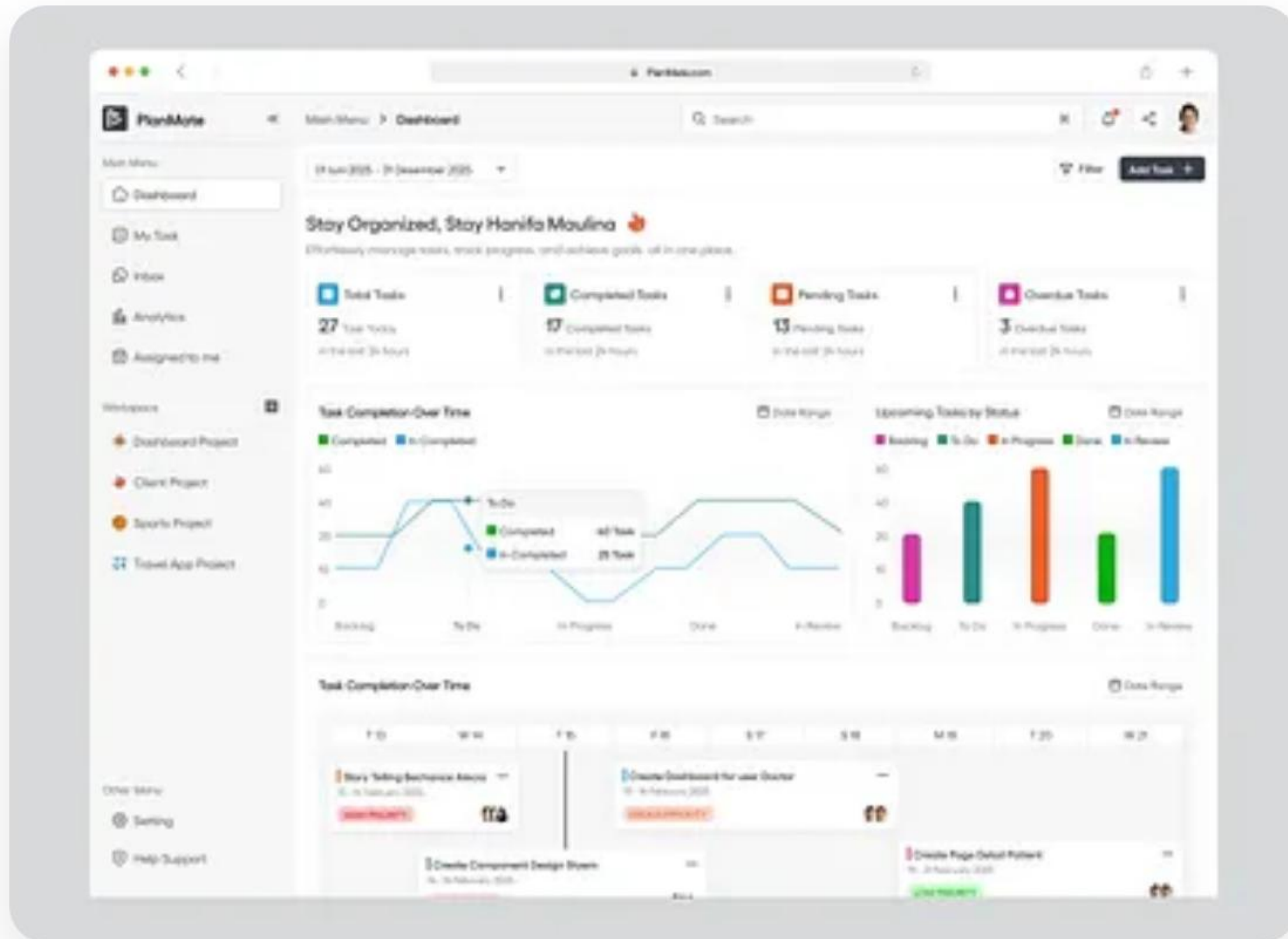
PaaS: Platform

Platform as a Service provides a framework for developers to create, test, and deploy applications without managing underlying hardware.

- Built-in development tools.
- Database management.
- Runtime environments.
- Examples: Heroku, Google App Engine.



SaaS: Software



Software as a Service allows users to connect to and use cloud-based apps over the Internet.

- No installation required.
- Automatic updates.
- Subscription-based (Opex).
- Examples: Gmail, Salesforce, Slack.

Public vs. Private Cloud



Public Cloud

Services provided over a network for public use. Owned by third-party providers (e.g., Azure, AWS).



Private Cloud

Cloud infrastructure operated solely for a single organization. It can be on-premise or managed externally.

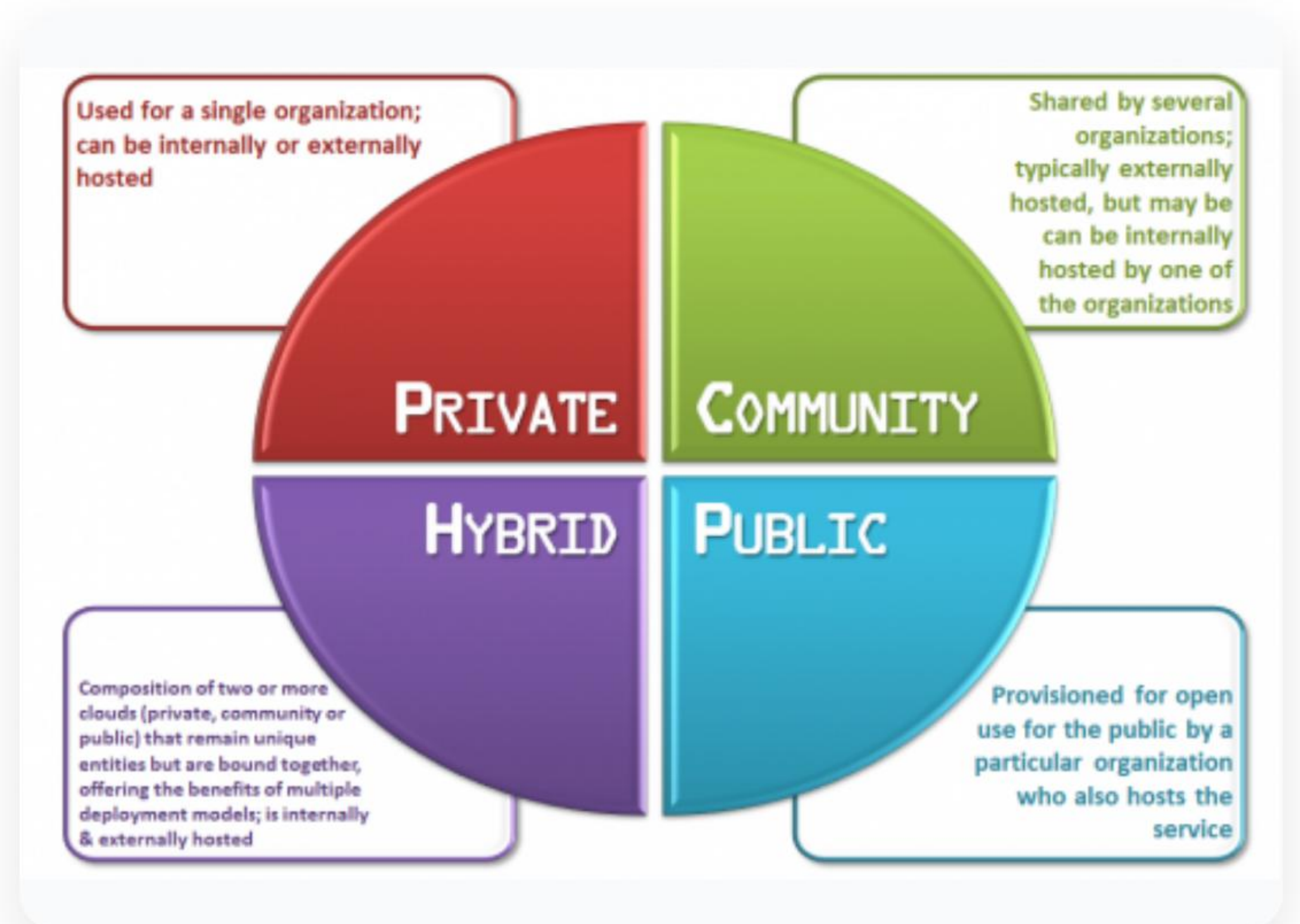
Community & Hybrid Models

Community Cloud

Shared by several organizations with shared concerns (e.g., mission, security, compliance).

Hybrid Cloud

Composition of two or more distinct clouds (Private, Community, or Public) that remain unique entities.



Cloud Management Roles

Entity	Function	Key Responsibility
Cloud Broker	Intermediary	Negotiates between users and providers.
Cloud Coordinator	Resource Manager	Assigns resources based on QoS/credits.
Cloud Exchange	Marketplace	Maps demand to available provider services.

Challenges & Barriers



While benefits like cost and agility are high, security remains the most significant hurdle for enterprise cloud adoption.

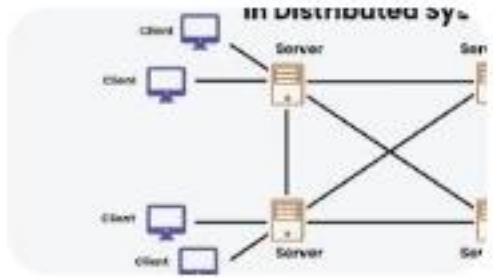
Summary

Cloud Computing paradigms represent the shift from local ownership to utility-based service consumption, driven by virtualization and global connectivity.

Questions?

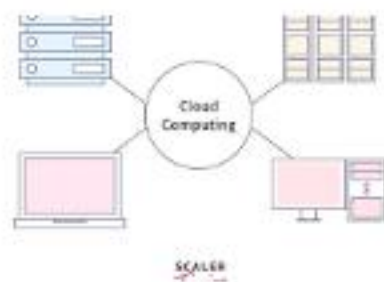
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